

CSS Layouts

Part of **Web Development Foundations** · Estimated time: 50 min

Introduction

CSS controls how HTML looks and where things sit on the page. Before touching layout, it helps to understand the box model: every element is a rectangular box made of content, padding, a border, and a margin, stacked from the inside out.

Core ideas

- The box model: padding sits inside the border (space around your content), margin sits outside it (space between elements). box-sizing: border-box makes width/height include padding and border, which is usually what you want.
- Flexbox (display: flex) is built for one-dimensional layouts — a row or a column of items. justify-content controls spacing along the main axis, align-items controls the cross axis.
- CSS Grid (display: grid) is built for two-dimensional layouts — rows and columns at once. grid-template-columns defines the column widths directly.
- Rule of thumb: reach for flexbox when arranging items in a line, grid when you're laying out a whole page or a card grid.

Example

```
.card-row {
  display: flex;
  gap: 16px;
  justify-content: space-between;
}
.card {
  flex: 1;
  padding: 16px;
  border: 1px solid #ddd;
}
```

Try it yourself

Build a row of three equal-width "cards" using flexbox, each with a heading and a short paragraph, evenly spaced with a gap between them. Then try converting the same layout to CSS grid with grid-template-columns: repeat(3, 1fr) and compare how much CSS each approach needed.

Further reading

- [MDN — CSS Layout](https://developer.mozilla.org/en-US/docs/Learn/CSS/CSS_layout)

https://developer.mozilla.org/en-US/docs/Learn/CSS/CSS_layout

- **CSS-Tricks — A Guide to Flexbox**

<https://css-tricks.com/snippets/css/a-guide-to-flexbox/>

ModuloTrainer reading material — return to the app to mark this module complete.